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Geocarto International
Publication details, including instructions for authors and subscription information:
http://www.tandfonline.com/loi/tgei20

Mapping Marine Habitats in Otago, Southern New Zealand

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Published online: 02 Jan 2008.

To cite this article: Jim Fyfe, Steven A. Israel, Albert Chong, Norhadi Ismail, Catriona L. Hurd & Keith Probert (1999) Mapping Marine Habitats in Otago, Southern New Zealand, Geocarto International, 14:3, 17-28, DOI: 10.1080/10106049908542113

To link to this article: http://dx.doi.org/10.1080/10106049908542113

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Abstract

Colour aerial photographs and field surveys were used to assess the extent and dynamics of giant bladder kelp, Macrocystis pyrifera, and eelgrass, Zostera novazelandica, beds. A common framework was developed for acquiring, integrating and analysing the spatial data. The two biological systems present different remote sensing and management problems. These differences help demonstrate the need for specific consideration of natural biological cycles and local management issues when developing remote sensing projects for marine ecosystems. A good local understanding of the spatial and temporal dynamics of biological systems is essential to ensure that environmental indicators monitored using remote sensing are meaningful and that appropriate standards are set.

Introduction

Most marine vegetation types occupy a zone adjacent to land too narrow to be mapped using remote sensing techniques. Those that can be remotely mapped and monitored include extensive seagrass beds (Kelly 1991), and kelp canopies (Jensen *et al*. 1980). Multidisciplinary research underway at the Otago University is using remote sensing techniques to assist study of the giant bladder kelp, *Macrocystis pyrifera* (Figure 1) near Pleasant River and eelgrass *Zostera novazelandica* (Figure 2) at Harwood in Otago Harbour (Figure 3).

Agenda 21, the comprehensive action plan adopted by the United Nations Conference on Environment and Development (UNCED) in 1992, encourages the identification and conservation of 'marine ecosystems exhibiting high levels of biodiversity and productivity'. *Macrocystis* and *Zostera* habitats are known worldwide for their diversity and production (North 1971, Zieman and Wetzel 1980). They provide shelter for many benthic species and can act as a nursery for fish (Carr 1989, DeMartini and Roberts 1990, Worthington *et al*. 1992). Despite this, there have been very few studies on the distribution, extent or ecology of *Macrocystis* and *Zostera* populations in New Zealand (Rapson *et al*. 1942, Hay l 990, Turner *et al*. 1996).

There have been many studies of *Macrocystis* beds in California where harvesting began early this century (North 1971, Foster and Schiel 1985, Deysher 1993). Annual productivity in these beds is 400-820g C/m^2/yr (Clendenning 1971, Coon 1982) and they are an important source of near-shore detritus (Gerard 1976, Duggins *et al*. 1989). The high productivity of *Macrocystis* has attracted the attention of New Zealand's aquaculture industry who are interested in harvesting it as a food source for abalone (*Haliotis iris*) (Bremner 1993a, Schiel *et al*. 1997). The Ministry of Fisheries require information on the ecological impact of removing seaweed biomass before harvesting will be allowed (Bremner 1993b). *Zostera* habitats also contribute large amounts of organic matter to coastal environments. For example, the annual productivity for *Zostera marina* in the northern Atlantic is 60-340g C/m^2/yr (Zieman and Wetzel 1980), most of which enters the detrital food chain (Nienhuis and Groenendijk 1986).

Information about energy transfer between habitats is needed in order to predict coastal ecosystem response to environmental change (Meulstee *et al*. 1986, ORC & DCC 1991). New Zealand's Resource Management Act (1992) requires managers to undertake resource monitoring, and

Figure 1 The floating canopy of *Macrocystis pyrifera* beds near Pleasant River.

Figure 2 Extensive *Zostera novazelandica* beds exposed at low tide in Harwood. Otago Harbour

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'indicator species' are being sought to provide representative information about environmental quality (MfE 1997). Remote sensing can be used to monitor marine habitats (Meulstee *et al*. 1986, Ferguson *et al*. 1993, North *et al*. 1993), but to distinguish anthropogenic effects requires considerable knowledge of natural changes and the human threats (North *et al*. 1993).

Large fluctuations in *Macrocystis pyrifera* canopy area and disappearance of entire beds occur naturally on the Californian coast (North *et al*. 1993). Nevertheless, long term declines as a result of turbidity caused by construction projects could be distinguished using photographs (North *et al*. 1993). Declining water quality has also caused the widespread demise of seagrasses (Cambridge and McComb 1984, Dennison *et al*. 1993). High nutrient levels were directly implicated when extensive mats of green algae smothered 10 ha of *Zostera* (den Hartog 1994). The cumulative effects of local impacts such as damage from mooring chains, reclamation and dredging also pose a threat to seagrasses (Cambridge and McComb 1984, Hastings *et al*. 1995).

Comparing *Macrocystis* and *Zostera* projects allowed us to explore different ways remote sensing can be used to gain information about marine habitats. To do this, we first considered biological and site characteristics in relation to image options available, as these are important for deciding on the classification scheme and overall remote sensing strategy (Congalton 1991). We identified thematic cover classes that can be distinguished from images and observed changes in cover over time. Our observations allow us to consider how remote sensing may assist with management of coastal habitats.

Biological considerations

Both seagrass and *Macrocystis* beds are recognised as dynamic habitats on a variety of spatial and temporal scales (Clarke and Kirkman 1989, Dayton *et al*. 1984, North *et al*. 1993). The biology of these habitat-forming species differs; however, *Macrocystis pyrifera* (Phaeophyta: Lessoniaceae) is a large brown alga with a microscopic gametophyte and a macroscopic sporophyte phase (Foster and Schiel 1985). The sporophytes can grow to the water surface from rocky reefs at depths of up to 20 m or more (Foster and Schiel 1985). Most *Macrocystis* biomass is at the surface (North 1972) in the form of a buoyant canopy of blades that have air-filled bladders at their base (Figure 4). Blades arise from fronds that are attached to the anchoring, perennial holdfast by narrow, flexible stipes (North 1972). *Zostera* (Magnoliophyta: Zosteraceae) grows in the sediment of comparatively sheltered harbours, inlets and estuaries. It has grass-like leaves and a dense system of rhizomes and roots, which anchor the plant to the substratum (Figure 5). *Zostera* is a flowering plant that forms dense meadows mainly by vegetative propagation (Olesen and Sand-Jensen 1994), though environmental factors can influence life-history strategies (van Lent and Verschuure 1994). Differences in population biology between *Macrocystis* and *Zostera* are expressed in spatial and temporal patterns of recruitment and bed development, thus influencing remote sensing resolution choices.

Site considerations

There is over 300 ha of *Macrocystis pyrifera* habitat adjacent to the 4 km stretch of coast south of Pleasant River (45° 35' S. 170° 44' E, Figure 3). This site was selected because the extensive beds were close enough to shore to allow landmarks to be included in photographs. *Macrosystis pyrifera* is the only seaweed that forms a surface canopy on these offshore beds, but the extent of this canopy is extremely variable over time. Sea level access to the *Macrocystis* beds requires a boat and changes in tide and weather make it difficult to compare canopy appearance between sites. Therefore, aerial photographs provide the best indication of spatial differences in canopy density, but diving surveys are required to investigate underlying population differences. Apart from the *Macrocystis* canopy, seawater is the major cover at the site.

The Harwood study site (45° 49' S. 170° 40' E, Figure 3) has the most extensive *Zostera* beds in Otago Harbour and a range of habitats and covertypes occur in association with these. Shell banks and beds of cockles (*Austrovenus stutchburyi*) are present. Harwood has good road access

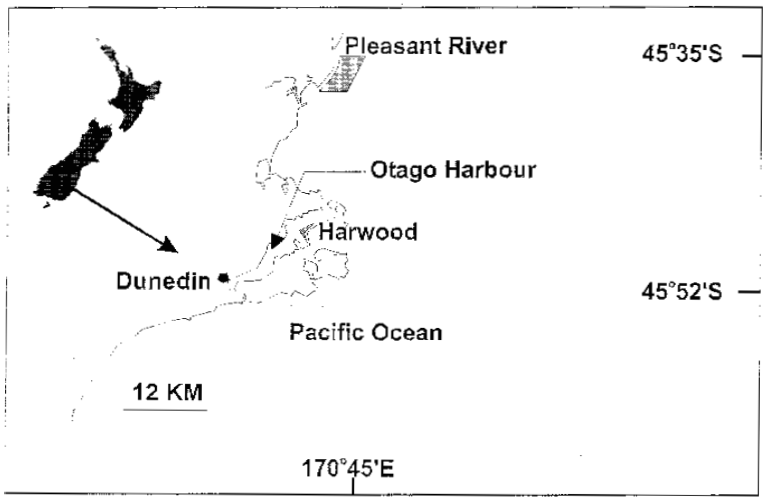

Figure 3 Map of the study sites, Otago, New Zealand.

Figure 4 *Macrocystis pyrifera* frond at the sea surface showing blades (A), bladders (B) and stipes (C), (photo width: 1m).

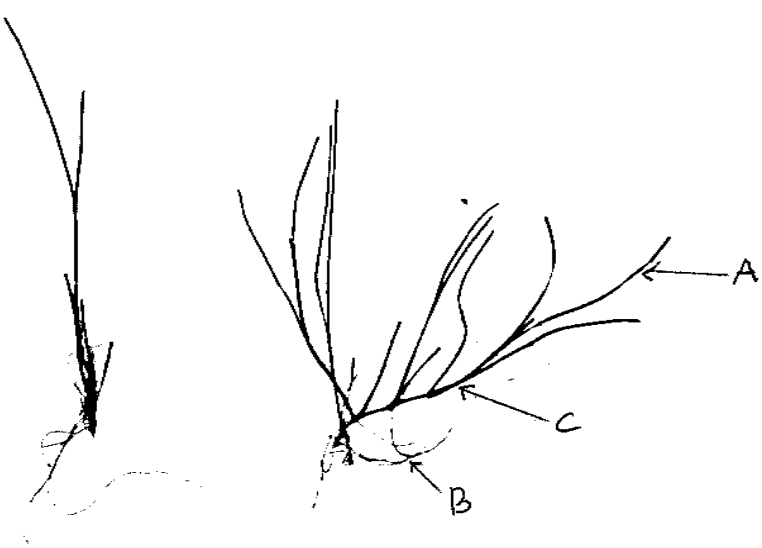

Figure 5 *Zostera novazelandica* plants (1/3 x original size) consisting of leaves (A), roots(B) and rhizomes(C).

and, at low tide, it is possible to view and compare different substrata and vegetation types. Shore level and sediment type can influence sediment water content and this can affect reflectance characteristics (Garono and Strittholt 1996). Sediment types may influence the distribution and percentage cover of *Zostera* in addition to influencing the spectral characteristics of *Zostera* classes. There was a good knowledge of the covertypes at this site from a previous study (Israel and Fyfe 1996). While no significant changes were observed in the distribution of *Zostera* at the Harwood site over the period of study, percentage cover of *Zostera* leaves and the macroalgal community did vary. The spectral characteristics of these covertypes need to be fully explored before any attempt is made to detect changes using classified maps.

Remote sensing options available

The remote sensing options for these projects were photography and SPOT imagery. Landsat data were not considered as no line-of-sight receiving station existed for New Zealand areas. The spectral region between 400 and 600 nm provides the only radiometric information about subsurface features deeper than a few metres (Austin 1974). Near infrared light can be useful for detecting *Macrocystis* canopy at the water surface (Jensen *et al.* 1980, Augenstein *et al.* 1991, Mouchot and Belsher 1991, Deysher 1993) or providing additional information about the density and biomass of seagrass (Meulstee *et al.* 1986). However, colour infrared photography was impractical as the film is processed overseas and SPOT data are too expensive for examining seasonal changes (Israel and Fyfe 1996). Using aerial photography enabled us to choose appropriate weather and sea conditions, tide and sun angle to ensure high quality images (Fyfe and Israel 1996).

Classification scheme

The information obtained from aerial photographs was considered in relation to the important thematic classes identified for each habitat and the extent of field verification possible. Despite similarities in data acquisition, different classification schemes were chosen for these remote sensing projects.

For the *Macrocystis* project the objective of classification was to investigate whether differences in canopy cover could be distinguished. Stratified sampling based on distinct classes can allow more efficient estimation of population characteristics than simple random sampling of the whole population (Barnett 1992). Classes have commonly been

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derived from photo-interpretation of algal density or spacing, and used as reference data to assess the accuracy of other remote sensing techniques to detect kelp (Augenstein *et al.* 1991, Deysher 1993). As canopy closure classes can be artificial, imposing discrete limits on a continuum, it is unlikely that such reference data are of the high accuracy needed for fair assessment (Congalton 1991). An unsupervised classification was chosen so that statistically based groupings of canopy spectral data could be examined in the field (Congalton 1991).

A hierarchical scheme was used to examine the *Zostera* habitat as there were many potentially useful classes visible in the imagery and these could be easily verified at the field site. This allows for categories in the initial supervised classification to be collapsed until the desired accuracy is achieved (Congalton 1991).

Methods

Aerial photography

Vertical photographs were obtained using a 60 X 90 mm format Mamiya camera with a focal length of 90 mm. Kodak 220 colour negative film and a UV filter were used. Photography for the purpose of mapping and monitoring Pleasant River *Macrocystis* beds began in April 1994. Photographs were obtained at least twice a year until April 1998. Flying height above the beds was approximately 3000 m, providing a 3 km field of view perpendicular to the coast. The flight path followed the coast and included 500 to 1000 m of land in each photograph. Since August 1995, *Zostera* beds at Harwood were photographed at an altitude of about 1800 m during the same flight.

Ground control points

Both projects used global positioning system (GPS) receivers to obtain ground coordinates for features on the adjacent land and for targets set up at the sites prior to the flights. White PVC plastic sheets (4 x 4 m) were placed on top of the *Macrocystis* canopy during a flight in April 1996 and 3 x 0.5 m plastic strips were pegged to the substratum in the *Zostera* beds prior to flights.

Image processing

Photographs for examination of cover types were acquired on 23 August 1995 for the *Macrocystis* site and on 22 April 1997 for the *Zostera* site. For both sites the aerial photographs were scanned to an approximate resolution of 1.5 m per pixel. These scanned images were transferred to ERDAS / Imagine for further processing. Images were registered to the New Zealand Map Grid (NZMG). Mosaics were constructed to include the areas of interest in a single image. The effects of vignetting (Lillesand and Kieffer, 1994) were minimised by using the centre portion of adjacent images.

Unsupervised classification

The unsupervised classification process was repeated a number of times for images of both sites in order to examine the influence of class number and image resolution. The resulting class information proved valuable for selection of reference areas at the *Zostera* site.

A six-class unsupervised classification of the *Macrocystis* site was selected for field verification. Two of these classes accounted for most of the *Macrocystis* area (Figure 6). Names were given to the two *Macrocystis* canopy classes based on observations at the field sites. Sites in the first class had a closed canopy (CC) with very few gaps. Sites within the second class had a broken canopy (BC), with many gaps. Only the offshore *Macrocystis* beds were classified, and the clear water adjacent to the outer beds required masking from the classification.

Ground data collection for *Macrocystis*

Several field surveys were undertaken in November and December 1995 to ground-truth the *Macrocystis* canopy covertypes indicated by the classification. Three sites were randomly selected in each canopy class and at each site. SCUBA was used to sample *Macrocystis* density along three 25 m transects. The maximum holdfast width and number of fronds were recorded for each alga along a transect. *Macrocystis* were collected to give an estimate of algal weight in different holdfast size groups. Using the same methodology, field-work was undertaken again in November and December 1996 to assess temporal changes within a class. Estimates of density were compared using *t*-tests. The mean and standard error for biomass estimates were calculated from density and weight information (Stuart and Ord 1994) for algae with a maximum holdfast width greater than 15 cm. The class biomass estimates were compared by calculating the difference between means and relating this to an estimate for the combined standard error.

Ground data collection for *Zostera*

Several field visits were undertaken during low tides in April - June 1997 to assign and then refine covertypes. Map co-ordinates giving northings and eastings in metres simplified the process of locating features at the site. The covertypes identified by Israel and Fyfe (1996) were used as a reference, but some additions and modifications were necessary. Twenty-three covertypes were identified. Percent ground cover of *Zostera* was measured using a photographed quadrat (Lanyon and Marsh 1995) at selected sites over the beds.

Supervised classification of the *Zostera* site

For each of the covertypes, six reference areas were collected. Due to the limited extent and localised nature of some of the covertypes, region-extracted sampling was applied (Campbell 1996). Each sample area contained about 70 pixels. These larger areas ensured that sufficient variation was acquired to fully characterise the covertype population. Three classification trials were performed, each using different reference data in order to minimise subjectivity and micro-climate effects.

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Accuracy assessment

Map accuracy was assessed using error matrices. For the *Macrocystis* canopy map (Figure 6), reference data derived from the randomly selected field sites were compared with the classification of corresponding spectral information from the image. As the focus in this case was to determine how distinct the canopy classes were, the water classes were grouped to form a single class in the error matrix. Reference data for water were derived from photo-interpretation of randomly generated points. For the *Zostera* map (Figure 8), data were divided into independent training and test areas and the accuracy of the classification of the known test areas was calculated. The supervised classification was performed 3 times using different training and test sites to examine differences between accuracy assessments.

Results

Observations of *Macrocystis* canopy changes

Photographs taken in April (autumn) were useful for distinguishing new *Macrocystis* canopy formed over summer while October (spring) photographs showed the extent of canopy damage by winter storms. After extensive removal of surface material at the Pleasant River site in August (winter) 1994, about 50 % of the *Macrocystis* beds reformed a dense canopy in February 1995. By August 1995 spatial differences were apparent within the new canopy area (Figure 8). These differences did not persist, and by April 1996 a dense canopy covered the entire area. By October 1996 the canopy formed in 1995 was distinct from that formed in 1996 as it was more broken or patchy.

Differences between canopy classes

The closed canopy and broken canopy classes, high-lighted in Figure 6, include most of the *Macrocystis* canopy area. Several distinct water masses, ranging from turbid to clear, accounted for the other four classes. One of these water classes included some low-density (usually peripheral) *Macrocystis* canopy, but this accounted for only a small area of the beds. Spectral data corresponding to the randomly selected field sites were used to generate an error matrix (Table 1).

Differences in density and biomass between *Macrocystis* canopy classes (Table 2) were not significant. The overall bed biomass (± standard error) for November 1995, calculated using separate classes was 7900 ± 1300 tonnes. If field information was combined to form a single

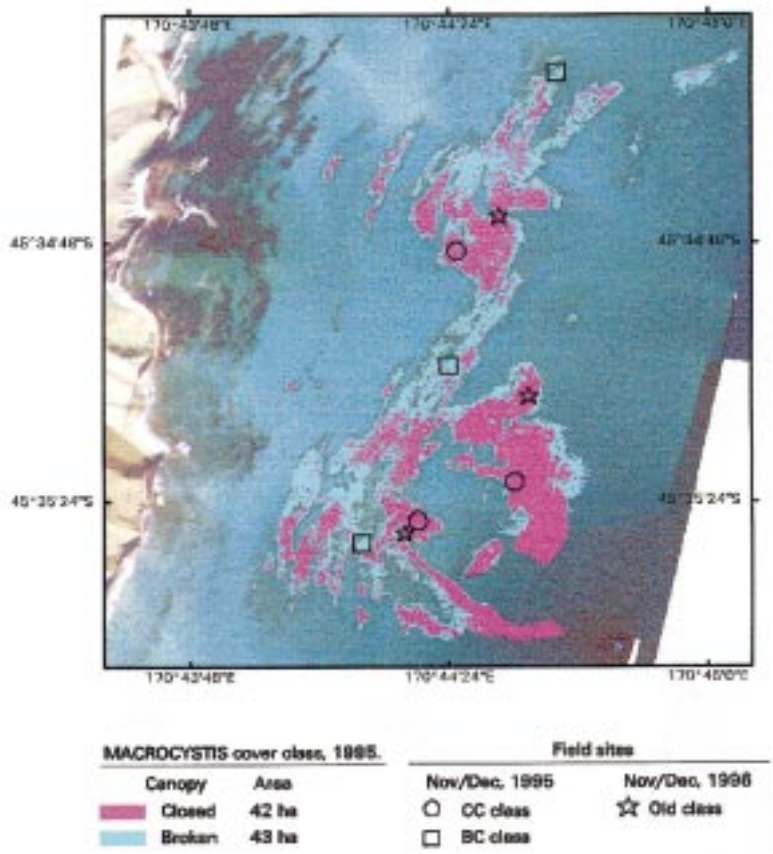

Figure 6 August 1995 photograph mosaic with the two *Macrocystis* classes (CC & BC) identified by overlaying the appropriate classes from a six class classification. Only the offshore beds were classified. Field sites are shown.

Table 1 Error matrix showing the omission and commission errors for samples of spectral data from the two *Macrocystis pyrifera* canopy classes and the surrounding seawater. Canopy class samples were based on the location of the randomly selected field sites. 120 pixels were randomly selected to provide representative spectral information for the combined water classes.

Covertype	CC	BC	Water	Total	User's accuracy (%)
Closed canopy (CC)	**147**	10	0	157	*93.6*
Broken canopy (BC)	9	**128**	10	147	*87.1*
Water	0	7	**110**	117	*94*
Total	156	145	120	**421**	
Producer's accuracy (%)	*94*	*88*	*91.6*		***91.4***

Table 2 Density and wet biomass estimates (mean ± one standard error) for the two *Macrocystis pyrifera* classes sampled in November 1995 (CC and BC), and for the CC class sampled in November 1996 (Old). Estimates are for algae with holdfast width greater than 15 cm.

	Density (algae/m^2)	(Biomass/m^2)
Closed canopy (CC)	1.16 ± 0.13	10639 ± 1566
Broken canopy (BC)	0.89 ± 0.09	7884 ± 1424
2 year old CC (Old)	0.41 ± 0.05	3761 ± 1237

Table 3 Major covertypes at Harwood intertidal area in April 1997.

Covertypes	*Percent cover*	*Area (ha.)*
Dense *Zostera*	70-100	25
Medium dense *Zostera*	30-70	20
Sparse *Zostera*	<30	33
Dense *Zostera*/ macroalgae	70-100	4
Total eelgrass area		**82**
Cockles/sand		57
Bare sand		30

'*Macrocystis*' class, the estimate was 8100 ± 1000 tonnes.

There was a significant reduction in *Macrocystis* density between November 1995 and November 1996 for the 1995 closed canopy class ($p < 0.006$). There was also strong evidence for biomass differences between the first and second year (difference/SE = 3.45).

Mapping Zostera *covertypes*

Zostera distribution was clearly visible in the mosaic of scanned photographs (Figure 7). Darker coloration was obvious in tidal pools or drainage depressions where *Zostera* and macroalgae occurred together in high densities. There was considerable contrast between the dense *Zostera* areas (Figure 9) and sandbars but less contrast between sparse *Zostera* (Figure 10) and sand or shell substrata. The map of covertype distribution predicted using a supervised classification procedure (Figure 8) corresponded closely with the *Zostera* beds identified in the photographs.

A supervised classification of the original data set produced 23 covertypes that were merged into the 6 major categories shown in Table 3. Covertypes lacking *Zostera* included bare sand and shelly sand (Table 3). Macroalgae were attached to the *Zostera* leaves or anchored to shell fragments. Based on this classification total *Zostera* bed area at Harwood was 82 ha, 33 ha (40%) of which was sparse *Zostera*.

Contingency matrices for the *Zostera* site are shown in Table 4. The overall accuracy for the training data determined from all three classifications was 94.6%. This describes how well training set pixels were classified and does not indicate the accuracy of the mapping (Lillesand and Kieffer, 1994). The mapping accuracy is shown in the error matrices on the test area. The overall classification accuracy determined from all three classifications was 90%. Minor confusion occurred between covertypes in each classification. User's accuracy indicating the likelihood of finding the classified covertypes *in situ*, was between 83 and 96%.

Table 4 Confusion matrix of training and test data sets of Harwood intertidal area using the result from supervised classification.

Training data

Covertypes	1	2	3	4	5	6		Total	User's Accuracy	
1	**848**	0	5	0	15	0		868	97.7	1 = Dense *Zostera*
2	1	**199**	0	0	0	0		200	99.5	2 = Dense *Zostera*/algae
3	12	0	**674**	17	1	1		705	95.6	3 = Medium *Zostera*
4	0	0	7	**1182**	38	33		1260	93.8	4 = Sparse *Zostera*
5	46	0	0	8	**1239**	9		1302	95.2	5 = Cockles/sand
6	0	0	0	41	20	**953**		1014	94.0	6 = Bare sand
							5095			
Total	097	199	686	1248	1313	996		**5349**		
Producer's Accuracy	93.5	100.0	98.3	94.7	94.4	95.7			**95.3**	

Test data

Covertypes	1	2	3	4	5	6		Total	User's Accuracy
1	**365**	0	24	1	10	0		400	91.3
2	6	**142**	0	0	0	0		148	95.9
3	10	0	**329**	53	0	0		392	83.9
4	1	0	9	**552**	20	19		601	91.8
5	52	0	2	8	**684**	31		777	88.0
6	0	0	0	79	18	**464**		561	82.7
							2536		
Total	434	142	364	693	732	514		**2879**	
Producer's Accuracy	84.1	100.0	90.4	79.7	93.4	90.3			**88.1**
						kappa = 0.852			

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Discussion

Image acquisition and processing

There were special requirements for photography at these sites. Fine, windless weather and low tide were prerequisites. Sun angle needed to be less than 60° to avoid the Sun's reflected glitter path entering the field of view (Austin 1974). If this occurred the influence on reflected radiance from water ruled out classification options.

Features at the study sites presented a variety of signatures in the aerial photographs depending on the percentage cover, species present, sediment type, and water clarity, as also reported by Ferguson *et al*. (1993). However, extraneous geometric effects on image brightness limit the potential for classifying photographic spectral information (Lillesand and Kiefer 1994). *Macrocystis* and clear seawater, for instance, both had low brightness values, so changes due to brightness fall-off and/or vignetting could influence their classification. It was possible to minimise the impact of these radiometric distortions by selecting photographs with the relevant information in a central position. Confusion still occurred between *Macrocystis* and clear seawater as a result of the similarity of their green reflectance values, and this required some offshore water to be masked from the classification process. There is a similar overlap in infrared reflectance values between *Macrocystis* and turbid water (Augenstein *et al*. 1991, Deysher 1993). Spectral characteristics of water masses within *Macrocystis* beds contribute to the canopy reflectance, particularly when there is a broken cover, so spatial autocorrelation of canopy data is inevitable.

There was some variability in brightness values between consecutive photographs perhaps due to minor changes in radiance associated with the proportion of different covertypes within a scene. This was most noticeable at the Harwood site where scene radiance changed markedly along the flight path. It was difficult to distinguish between medium dense *Zostera* and some cockle areas in the image. Variation in reflectance value within the same category of *Zostera* was also common. Differences in sediment wetness, or in *Zostera* leaf length, shoot density and the abundance of associated epiphytes could contribute to this. The frequent field visits to obtain ground-truth data were essential for selecting the appropriate field classes to minimise confusion while maintaining class utility.

Fieldwork

It took three days of intensive field work, requiring two boats and eight divers, just to compare two *Macrocystis* classes using methods that could provide adequate statistical information about the density and size of algae in these different canopy classes. It was not always possible to conduct the fieldwork immediately after aerial photographs were obtained due to changing weather conditions. Given the high growth rates of *Macrocystis* (Clendenning 1971) and the dramatic changes in the extent of beds, time delays could affect biomass estimates. In contrast *Zostera*, which could be accessed easily at low tide, did not undergo the rapid changes that made synchronous data collection so important. It was possible to fully explore the *Zostera* area and sample fixed sites on a regular basis.

Accuracy assessment

There is a natural break in the spectral data relating to the *Macrocystis* canopy as indicated by the unsupervised classification (Table 1). Verification of canopy type was restricted to the randomly selected field sites in each class, so small clusters were acquired from the corresponding areas of the image to produce an error matrix. A greater number of field sites could have given a better indication of accuracy (Congalton 1991, Ferguson *et al*. 1993), but was not possible due to the difficult access to the Pleasant River site. It was possible to distinguish a number of useful *Zostera* classes while maintaining high accuracy (Table 4). There was easy access to the Harwood site, but data collected for assigning covertypes were considered adequate to provide an indication of the accuracy. Random or systematic sampling of single pixels was unlikely to improve the accuracy assessment due to the localised nature of some covertypes and uncertainty about the match between field and pixel location at 1.5 m resolution.

Ecological monitoring

Most of the biomass of *Macrocystis pyrifera* is in the canopy (North 1972, Lobban 1978), and infrared reflectance values have been related to canopy biomass (Mouchot and Belsher 1991). The *Macrocystis* classes observed in this study represented canopy differences (Table 1), but differences in biomass associated with the classes were insignificant and did not improve overall bed biomass estimates. Canopy differences can arise for a number of reasons including the population age structure (Gerard 1984), substratum suitability or site exposure (Graham *et al*. 1997). The classes therefore do not represent useful information that can be generalised to give a better understanding of bed processes and available biomass. In contrast, classes identified from aerial photographs of *Zostera* beds can greatly assist ecological studies. The sparse class, for instance, is distinct in terms of above-ground material and can be considered in relation to substratum classes and shore level; information that is also available from an image.

The dynamics of ecological systems have important implications for the strategies of biological populations as well as the strategies for observing them. Storms have a considerable influence on *Macrocystis* beds (Figure 11), disrupting large areas of canopy at the same time thereby synchronizing recruitment (Seymour *et al*. 1989, Graham *et al*. 1997). *Macrocystis* is resilient under a regime of frequent disturbance (Dayton 1984) and population cycles are well documented (Rosenthal *et al*. 1974, Dayton *et al*. 1984, Dayton *et al*. 1992, Graham *et al*. 1997). In fact, there is only sufficient light for the growth of young sporophytes once the surface canopy has been substantially removed

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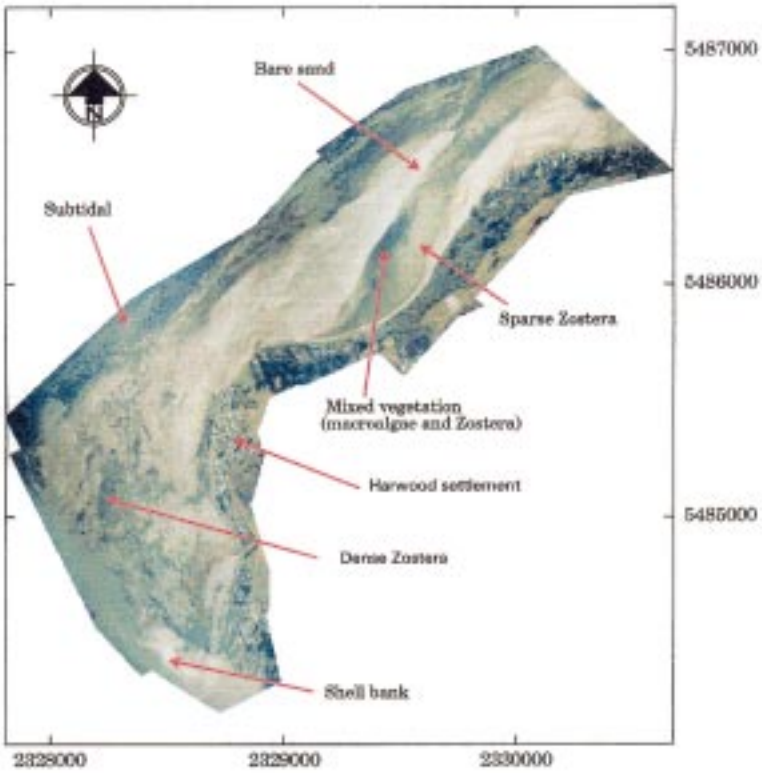

Figure 7 An aerial photograph mosaic of Harwood intertidal area in April 1997 showing the eelgrass and substrata distribution. The coordinate system is the New Zealand Map Grid; each grid represents 1000 m on the ground.

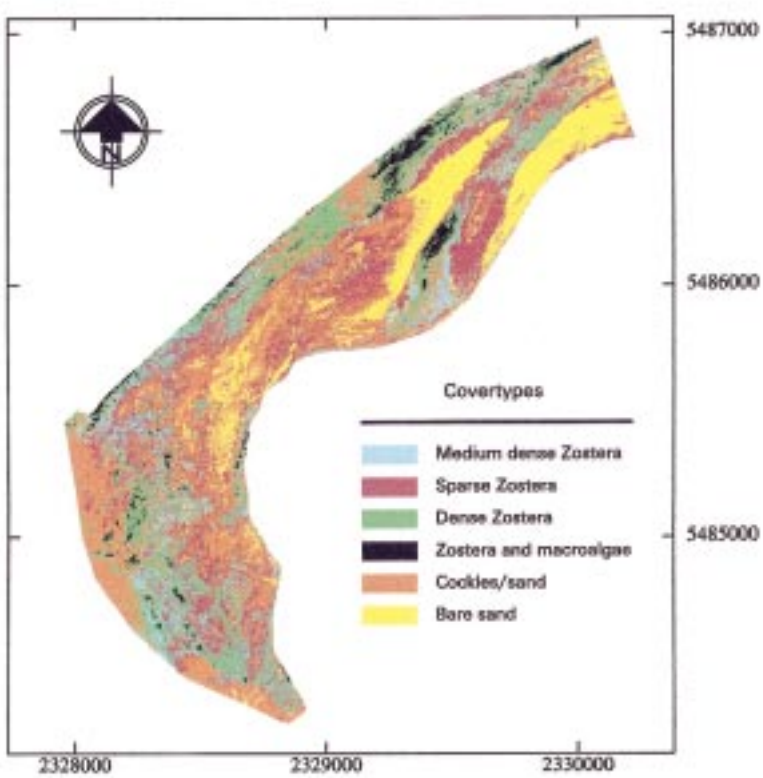

Figure 8 Distribution of major covertypes for the Harwood intertidal area in April 1997, produced using supervised classification.

(Reed and Foster 1984, Deysher and Dean 1986). Disturbance of *Zostera* beds is of a different nature and scale. 'Blow-outs' have been well described (Clarke and Kirkman 1989), but these are on a scale of 10's of metres rather than the 10's of hectares that can be disrupted in a *Macrocystis* bed during a storm. The rhizome and root system of *Zostera* mean that once plants have established they stabilise the sediment, preventing erosion (Armiger 1964). Disturbance events are likely to be localised unless they involve some aspect of water quality (Cambridge and McComb 1984) or disease (Short *et al*. 1988).

The high spatial resolution provided by photography enables the precise and accurate mapping necessary to study the dynamics of seagrass-associated bedforms (Clarke and Kirkman 1989). Sparse *Zostera* and bare sand classes obtained from classifying these photographs were more distinct (Table 3) than similar classes observed using SPOT imagery (Israel and Fyfe, 1996). Since sparse *Zostera* is an extensive and important class, photography may provide the best remote sensing option to investigate seasonal changes in the percentage cover of *Zostera*. However, a thorough field based accuracy assessment is needed to demonstrate the reliability of maps generated in this way.

Changes in *Macrocystis* canopy distribution over time can be observed easily and are of a scale that can be quantified using lower resolution images. Canopy development and demise appeared to be synchronised for the recruitment cohort that emerged in 1995. The reduction in mean density of *Macrocystis* between November 1995 and November 1996 (Table 2) was accompanied by a decline in surface canopy and biomass (Figure 12). Cohort age, deduced from changes observed using remotely sensed images, may therefore provide information about population biomass. The major advantage of photography for monitoring *Macrocystis* beds is the high temporal resolution it allows. As confusion between *Macrocystis* classes and clear water required some offshore areas to be masked from the classification process, digitising the perimeter of surface canopy is likely to provide the most reliable map of its extent.

Monitoring for management

Large changes in the extent of *Macrocystis* canopy observed during this study are similar to those recorded for beds elsewhere (e.g. Ebeling *et al*. l985, Dayton *et al*. 1992, North *et al*. 1993, Graham *et al*. 1997). This natural variability can make it difficult to establish whether human

activities have an impact on *Macrocystis* populations (North *et al*. l 993). The impact of poor water quality often involves small areas and sparse beds (Deysher 1993, Schiel *et al*. 1997), so it is unlikely that remote sensing of *Macrocystis* will give a good indication of ecosystem health on a regional scale. Despite this, ongoing monitoring of canopy cycles will provide important information for managers interested in either exploiting or protecting this productive coastal habitat (Foster and Schiel 1985, Bushing 1995, Graham *et al*. 1997).

Mapping the distribution of *Zostera* beds can provide an environmental indicator for harbour, inlet and estuarine systems. Only small changes in distribution were observed in this study. A large loss in the area of *Zostera* at Harwood is undesirable in terms of Agenda 21, and would almost certainly be an indication of more widespread environmental degradation. The methodology to observe and display spatial change in seagrass is an important prerequisite to detecting nation-wide changes (Ferguson *et al*. 1993). Annual photographs of significant *Zostera* beds, obtained at a time when plants have maximum surface cover, could provide the information necessary for state of the environment reporting. High resolution is not crucial for this monitoring, so SPOT imagery should be considered as it provides good regional coverage and the spectral information is more uniform throughout the image. Environmental standards set could be based on maintaining the area of *Zostera* beds in a region. When specific coastal developments threaten *Zostera* beds, local authorities could require the developer to monitor changes using high resolution photography (Bamber 1990).

Figure 9 Dense *Zostera novazelandica* at a permanent monitoring site, Harwood, Otago Harbour. The peg is about 25 cm high.

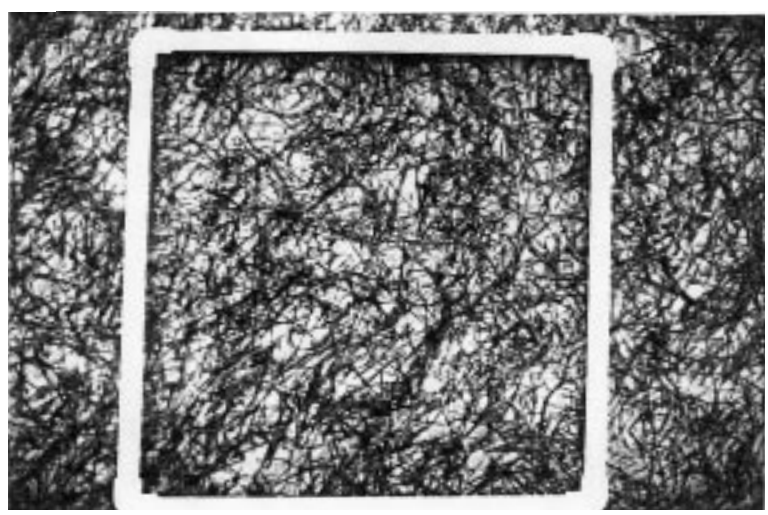

Figure 10 Sparse *Zostera novazelandica* consisting of narrow leaves and short shoots in a 0.25 m^2 quadrat.

Conclusions

Although *Macrocystis* and *Zostera* beds can be photographed in the same way, striking differences in biological cycles and field access create the need for different strategies to identify the classes or cohorts significant to ecology. The proximity of human populations presents a threat to *Zostera* beds in the form of harbour development or increased nutrient loading. This makes *Zostera* an ideal choice as an indicator for state of the environment reporting. The main threat to *Macrocystis*, on the other hand, is harvesting. While better understanding of our natural habitats is desirable, in this case further research may expose the beds to an increased interest in their exploitation. A long-term baseline study of the dynamics of non-harvested *Macrocystis* beds will be necessary to assess any impacts that harvesting has on production cycles.

While photographs have some processing limitations, they do allow low-cost surveillance and, therefore, have an important role in identifying habitat dynamics at a variety of scales. Those thematic classes that are important to biology and management should be identified as early as possible so that fieldwork can be conducted efficiently and other remote sensing options can be considered.

Acknowledgements

The authors thank staff and students of the Marine Science, Botany and Surveying Departments, Otago University, for their assistance in these projects. Special thanks to Roger Carmen, Maja Fricke, Jana Moeller and Jaqueline van der Pol who assisted in obtaining ground co-ordinates and creating the georeferenced base maps. A Department of Conservation contract provided funding for the initial mapping of

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Figure 11 Storms reduce *Macrocystis pyrifera* biomass and can have a considerable influence on population structure at the Pleasant River site.

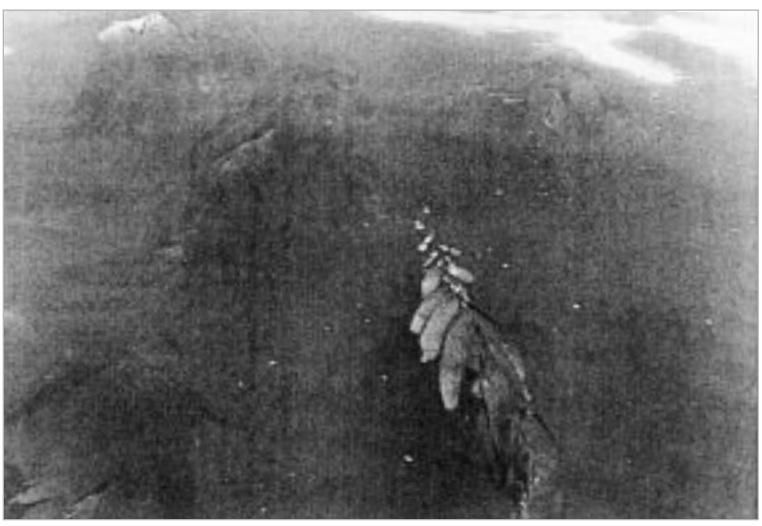

Figure 12 The canopy of these two-year old *Macrocystis pyrifera* was significantly reduced by storms during winter in 1996, allowing new recruitment (photo width: approximately 5 m).

Zostera at Harwood, reported in Israel and Fyfe (1996). Ongoing monitoring was made possible by a grant from the Otago University Division of Sciences.

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